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Kaggle Link: [PHEMS Hackathon: Pediatric Sepsis Prediction | Kaggle](#)

The Problem:

Sepsis in children admitted to the pediatric intensive care unit (PICU) is the leading causes of death in Europe. Early detection remains a difficult problem to solve, and every hour of delayed treatment means a 4-8% rise in mortality.

The goal of this Kaggle hackathon is to develop a machine learning algorithm that can provide the likelihood of sepsis in the patient, allowing for potential life-saving measures to be taken, while also reducing the strain on the healthcare system in Europe.

Data:

The data came from the PICU of Hospital Sant Joan de Deu and anonymized. Our output feature needs to have the ID of the patient, the time the prediction is made, and the probability of sepsis. The predictor data consist of several sections that happens at a specific time: device used, drugs given to patient, measurement of drugs given to patient, observations recorded during their stay, demographic, and procedures performed to a patient. Below are the sizes of each section of the dataset:

devices.csv:	751K
drugexposure.csv:	185K
measurement_lab.csv:	69.3K
measurement_meds.csv:	258K
measurement_observation.csv:	50.6K
observation.csv:	3807
person_demographics_episode.csv:	3391
proceduresoccurrences.csv:	771K

Task:

This is a classification problem that provides the probability for 2 separate classes: sepsis and not sepsis.

Proposed Methods:

Our approach will follow four main phases: feature engineering, dimensionality reduction, class imbalance handling, and classification.

Feature Engineering: We will create meaningful features from the raw clinical time-series data, such as rolling averages, trends in vital signs, and rate-of-change indicators. Missing data patterns will also be captured as features, since the absence of certain measurements can be clinically relevant.

Dimensionality Reduction: PCA will be used to reduce the number of features and remove redundancy among correlated variables, helping the models generalize better.

Class Imbalance Handling: Since sepsis cases are relatively rare in the dataset, we will apply techniques such as SMOTE or class weighting to ensure the model doesn't simply learn to predict the majority class.

Classification: We plan to evaluate KNN as a baseline and XGBoost as our primary model, given its strong track record with tabular clinical data and its ability to handle missing values natively. Hyperparameters will be tuned using cross validation.

Evaluation: Model performance will be measured using AUC-ROC and precision-recall metrics, with stratified k-fold cross-validation to ensure reliable results.

Anticipated Outcome: We expect to produce a model that can flag at-risk patients earlier than current clinical practice, with XGBoost likely being the stronger performer. The final model will include interpretable feature importances to support clinical decision-making.